back. Then it slowly backs up the side of the nest, balancing the egg between its wing-stubs, and topples the egg out on to the ground. It does the same with all the other eggs, until it has the nest, and therefore the attention of its foster parents, entirely to itself.

One of the most remarkable facts I have learned in the past year was reported from Spain by F. Alvarez, L. Arias de Reyna, and H. Segura. They were investigating the ability of potential foster parents—potential victims of cuckoos—to detect intruders, cuckoo eggs or chicks. In the course of their experiments they had occasion to introduce into magpie nests the eggs and chicks of cuckoos, and, for comparison, eggs and chicks of other species such as swallows. On one occasion they introduced a baby swallow into a magpie's nest. The next day they noticed one of the magpie eggs lying on the ground under the nest. It had not broken, so they picked it up, replaced it, and watched. What they saw is utterly remarkable. The baby swallow, behaving exactly as if it was a baby cuckoo, threw the egg out. They replaced the egg again, and exactly the same thing happened. The baby swallow used the cuckoo method of balancing the egg on its back between its wing-stubs, and walking backwards up the side of the nest until the egg toppled out.

Perhaps wisely, Alvarez and his colleagues made no attempt to explain their astonishing observation. How could such behaviour evolve in the swallow gene pool? It must correspond to something in the normal life of a swallow. Baby swallows are not accustomed to finding themselves in magpie nests. They are never normally found in any nest except their own. Could the behaviour represent an evolved anti-cuckoo adaptation? Has the natural selection been favouring a policy of counter-attack in the swallow gene pool, genes for hitting the cuckoo with his own weapons? It seems to be a fact that swallows' nests are not normally parasitized by cuckoos. Perhaps this is why. According to this theory, the magpie eggs of the experiment would be incidentally getting the same treatment, perhaps because, like cuckoo eggs, they are bigger than swallow eggs. But if baby swallows can tell the difference between a large egg and a normal swallow egg, surely the mother should be able to as well. In this case why is it not the mother who ejects the cuckoo egg, since it would be so much easier for her to do so than the baby? The same objection applies to the theory that the baby swallow's behaviour normally functions to remove addled eggs or other debris from the nest. Once again, this task could be—and is—performed better by the parent.